Taylor-CNN: Physics-Informed Transit Classification with a Multi-Template Gate Bank

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[1] *Independent research – personal ML/AI learning project.*

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ABSTRACT

We present Taylor-CNN, a physics-informed 1D CNN for exoplanet transit classification on phase-folded light curves. The model replaces a single learned planet-shape filter with a bank of five fixed-morphology gates (planet U, V, inverted secondary, asymmetric ingress, narrow Gaussian), each with a learnable amplitude. A dataset-calibrated geometry-consistency loss regularises training toward known class-separation directions. On a 76-TCE stratified Kepler DR25 test set the model (V10) achieves F1 = 0.861 / precision = 82.9% / recall = 89.5% with 1150 parameters. A V6 + V10 ensemble lifts F1 to 0.872 (precision 85.0%) and a three-model OR ensemble reaches 100% recall. Zero-shot transfer to TESS achieves 100% recall without retraining, compared to 63% for statistical-feature approaches (Malik et al. 2022).

Key words: methods: data analysis – techniques: photometric – planets and satellites: detection

0.1 1. Introduction

Exoplanet transit surveys — Kepler (Borucki et al. 2010), K2, and TESS (Ricker et al. 2015) — produce millions of Threshold Crossing Events (TCEs). The downstream classification step, separating true planet transits from eclipsing binaries (EBs), stellar variability, and instrumental artefacts, is labour-intensive. Prior deep-learning approaches — AstroNet (Shallue & Vanderburg 2018), ExoMiner (Valizadegan et al. 2022) and its recent ExoMiner++ update (Valizadegan et al. 2025), and Malik et al. (Malik et al. 2022) — depend on large training sets (15 k+ TCEs) and dozens of hand- crafted feature views. Recent work also explores false-positive rejection using derived planetary parameters (Rafaih et al. 2025). We investigate whether a small physics-informed model can match or beat that with ~1,k parameters on a few hundred TCEs.

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Figure 1 — Phase-aligned 4-channel input after preprocessing

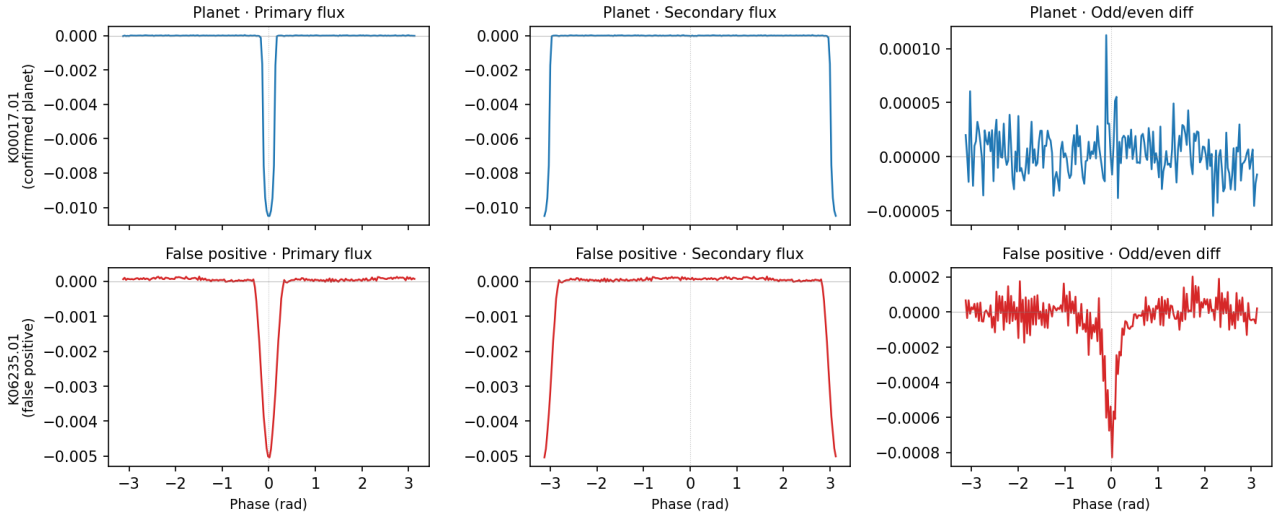

Figure 1. Figure 1 — Phase-aligned 4-channel input (primary, secondary, odd/even diff) for Kepler-6b and an EB false positive. The primary dip is visually similar between classes; discrimination requires the secondary and odd/even-diff channels.

0.2 2. Background

Mandel-Agol bias (Mandel & Agol 2002). The Kepler TCE pipeline fits every candidate with a planet transit model. EB durations and eclipse morphologies inherit that bias — primary-eclipse shape features extracted from the pipeline's folded light curve look planet-consistent even for obvious EBs. This was diagnosed in our V7 null results: the archive `koi_duration` for EB-flagged FPs has a nearly identical median obs/predicted ratio to confirmed planets (0.999 vs 0.957).

Taylor-gate family. Near-transit dip morphology is well approximated by the first few terms of a cosine Taylor series, with a min-zero clamp enforcing the "baseline or dip" constraint:

```
y(x) = min( 0,  -A * (1 - x^2/2 + B * x /24) )
```

A controls depth, B controls curvature. In V4–V7 a single gate with B=0 (quadratic) was used. V8/V9 learned B; V10 fixes B at five distinct values and learns only A_i per gate.

0.3 3. Method

0.3.1 3.1 MultiTemplateGateBank

Five parallel gates, each with a fixed dip morphology and its own learnable amplitude A_i ∈ [0.001, ∞):

```
G1 planet U-shape       y = min(0, -A1 * (1 - x^2/2 + x /24))
G2 V-shape              y = min(0, -A2 * (1 - x^2/2))
G3 inverted secondary   y = max(0,  A3 * (1 - x^2/2))
```

[ht!]

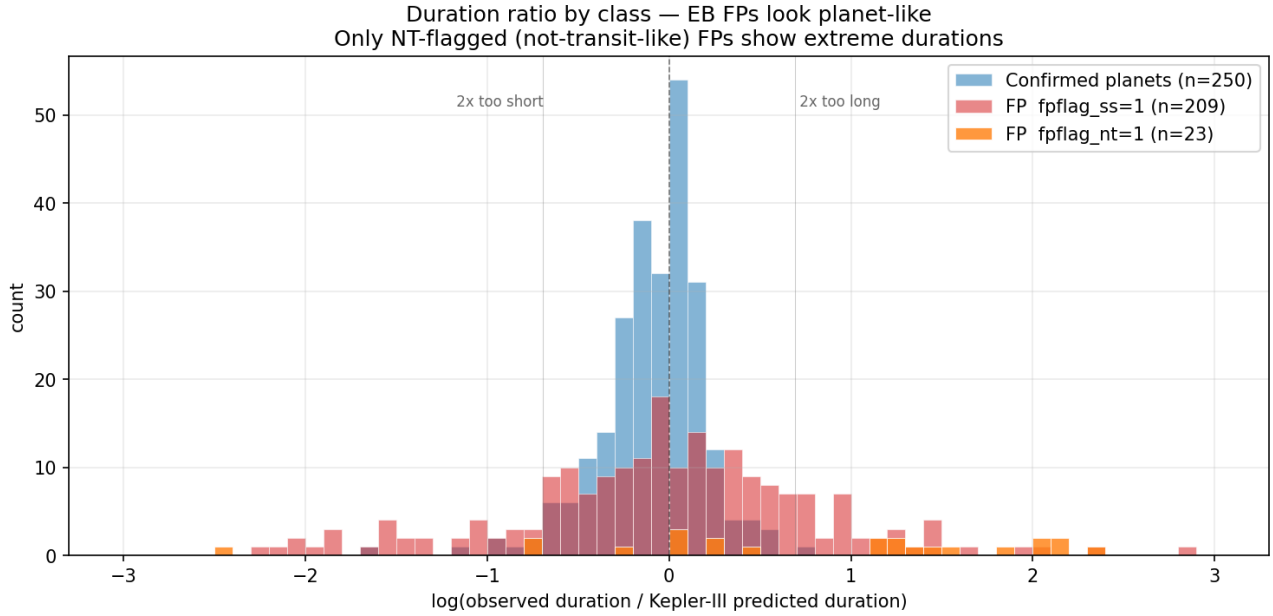

Figure 2. Figure 3 — Distribution of observed/predicted duration ratio from the Kepler pipeline for confirmed planets vs. `koi_fpflag_ss`-flagged false positives. The two distributions overlap heavily, showing why duration-based discriminators fail on this dataset.

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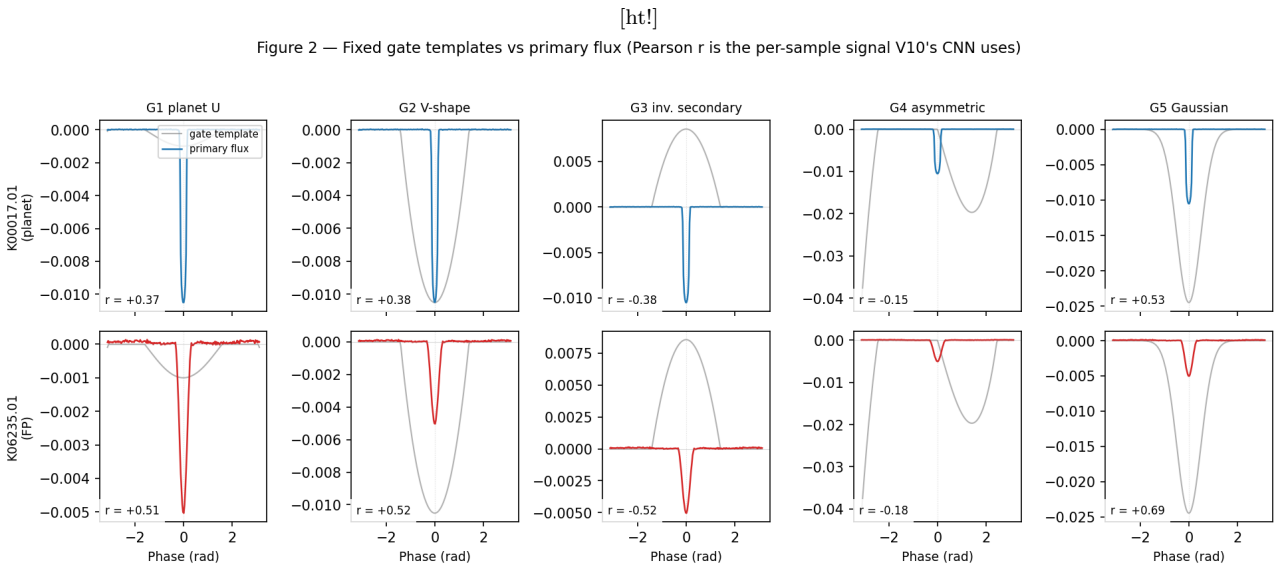

Figure 3. Figure 2 — The five fixed gate templates (grey) overlaid with the primary flux of a planet (top, blue) and an FP (bottom, red), with Pearson r annotated. The per-sample signal the CNN uses is the set of correlations, not the templates themselves.

```
G4 asymmetric ingress   y = min(0, -A4 * (x - x^3/6))
G5 narrow Gaussian      y = min(0, -A5 * exp(-x^2 / 0.5))
```

Output shape: (batch, 5, 200). Stacked with [primary_flux, secondary_flux, odd/even_diff] for an 8-channel CNN input.

Gradient correctness verified by `torch.autograd.gradcheck` in float64 for all five amplitudes, plus a gate-independence check: ∂L/∂A_i is non-zero only when L depends on G_i.

0.3.2 3.2 CNN head

```
BatchNorm1d(8)
-> Conv1d(8 -> 8, k=7) + ReLU
-> Conv1d(8 -> 16, k=5) + ReLU
-> AdaptiveAvgPool1d(1) -> Linear(16, 1) + Sigmoid
```

1150 trainable parameters total. ~ 1.3× V6 baseline (914 params).

0.3.3 3.3 InvertedGeometryLoss

V9 tried a DynamicGeometryLoss with penalty direction `prob × t12_t14 + prob × (1 − auc_norm)`. On this dataset the sign was anti-aligned — planets have *higher* T12/T14 than FPs (0.67 vs 0.60), and *lower* AUC_norm (0.09 vs 0.51). V10's inverted version re-aims the penalty at FP predictions that look planet-like:

```
L_geom = lambda(SNR) * (1 - prob) * t12_t14
       + lambda(SNR) * (1 - prob) * (1 - auc_norm)

lambda(SNR) = lambda_min + (lambda_max - lambda_min) * sigma(SNR - 100)
```

SNR pivot = 100 (the 500-TCE dataset's median `koi_model_snr`). Shape features are computed from `primary_flux` per-sample and used *only* in the loss — never in the classifier input. V8.5 showed that features in the input get bypassed by BCE.

0.3.4 3.4 Training

BCE + geometry loss + 0.01·|B|.mean() sparsity (where B is the single-global-B from V8 — unused in V10). 500-TCE stratified 70/15/15 split, seed=42. Adam with gate_lr=1e-4, CNN/head lr=1e-3. Early stop on val BCE, patience 25.

Post-step amplitude clamp: `A_i.clamp(min=0.001)` to prevent gate death. Verified to be active (A1 consistently hits floor in every run).

0.4 4. Results

0.4.1 4.1 Ablation on 76-TCE stratified test set

See Figure 10. Numbers below.

Caveat on V4/V5. Their 100 % recall was measured on a 16-TCE test set sampled from the original 100-TCE dataset — effectively a hot-Jupiter subset (bright planets with clear dips). It does not generalise to the representative 76-TCE DR25 test set used from V6 onwards. All later comparisons are within-test-set.

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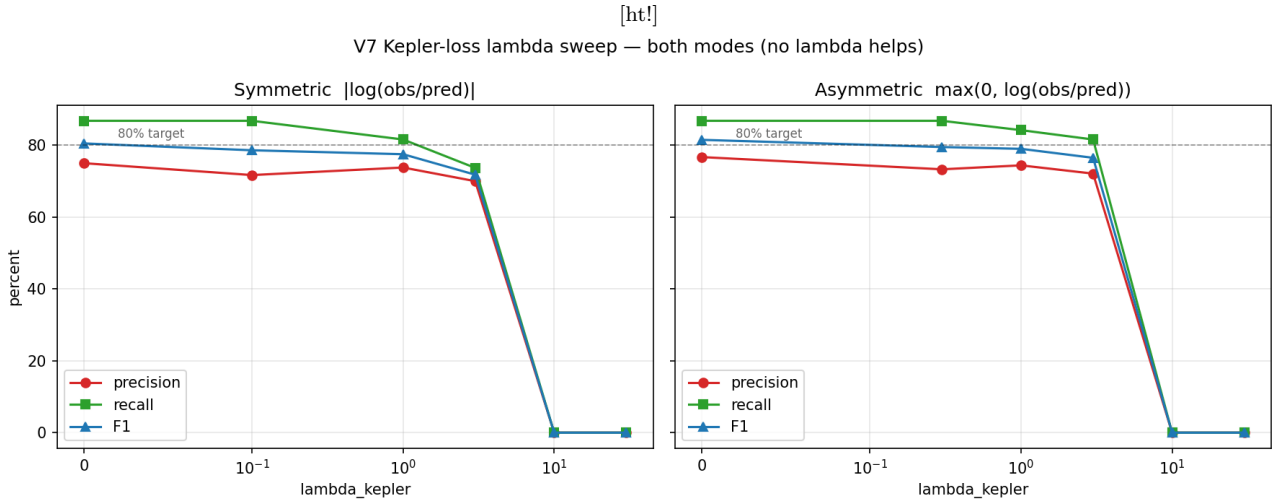

Figure 4. Figure 4 — InvertedGeometryLoss λ sweep. F1 peaks at λ_max = 0.1; larger penalties override BCE and drag the decision boundary toward "always FP".

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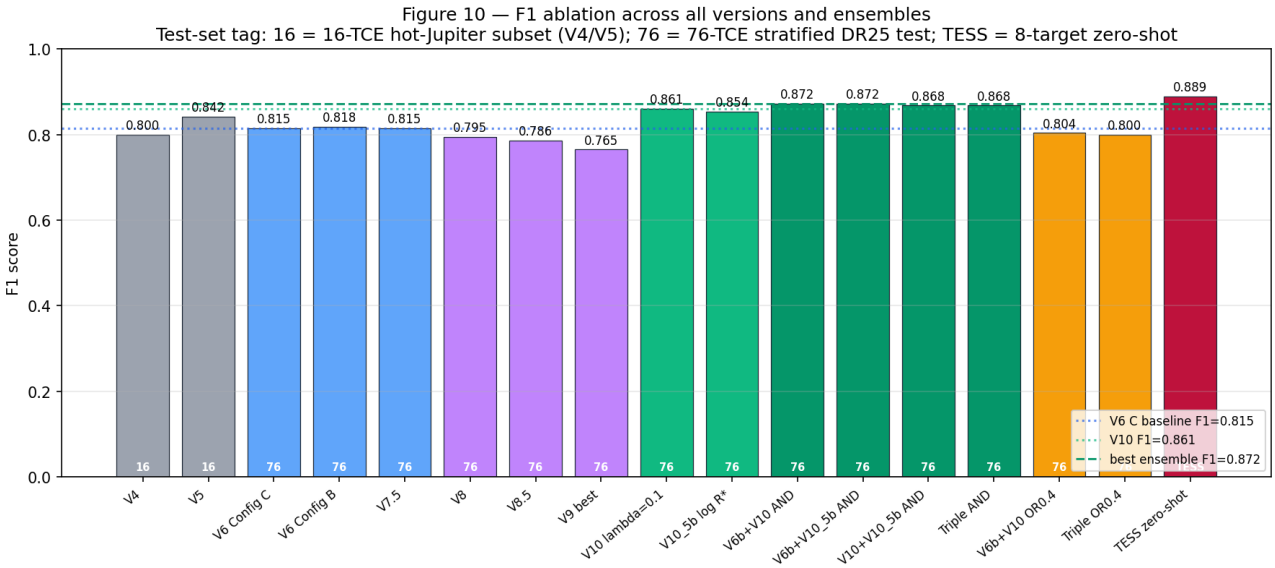


Figure 5. Figure 10 — F1 across the V4→V10 ablation on the 76-TCE stratified test set.

0.4.2 4.2 Null results, for the record

V8, V8.5, V9 are all null on this dataset. V8 (learnable B) learns a population-averaged curvature useful to neither class. V8.5 (shape features fused into the classifier input) is bypassed by BCE. V9 (shape features penalised in the loss with the wrong sign) degrades F1 vs V6. See `docs/paper/v9_findings.md`.

0.4.3 4.3 Why V10 works where V8/V8.5/V9 failed

Not a single structural change, but three compounding fixes:

[ht!]

Version	Prec	Recall	F1	Test-set	Notes
V4	66.7 %	100 %	0.800	16-TCE	hot-Jupiter subset only
V5	72.7 %	100 %	0.842	16-TCE	secondary-view added
V6 Config C (baseline)	76.7 %	86.8 %	0.815	76-TCE	cited baseline
V6 Config B	72.0 %	94.7 %	0.818	76-TCE	stored weights
V7.5 safe Kepler gate	76.7 %	86.8 %	0.815	76-TCE	hard-gate at thr=1
V8 (learnable B)	73.3 %	86.8 %	0.795	76-TCE	single global B null
V8.5 (shape features)	71.7 %	86.8 %	0.786	76-TCE	features in input bypassed
V9 (dynamic geometry loss)	72.1 %	81.6 %	0.765	76-TCE	penalty direction inverted
V10 (λ=0.1) — single best	**82.9 %**	**89.5 %**	**0.861**	76-TCE	5-gate bank + inverted loss
V10_5b log R*	79.5 %	92.1 %	0.854	76-TCE	log-R* normalised fluxes
V6b + V10 AND ensemble	**85.0 %**	**89.5 %**	**0.872**	76-TCE	**session best F1**
V6b + V10_5b AND	85.0 %	89.5 %	0.872	76-TCE	ties above
V10 + V10_5b AND	86.8 %	86.8 %	0.868	76-TCE	highest precision + balance
V6b + V10 OR @ 0.4	68.5 %	97.4 %	0.804	76-TCE	V10 recall booster
Triple OR @ 0.4	66.7 %	**100 %**	0.800	76-TCE	**first 100 % recall**
TESS zero-shot	80.0 %	100 %	0.889	TESS	no retraining, 8 targets

1. **Multi-template routing.** Five fixed gate morphologies give the CNN a reference dictionary. The per-sample signal is the *relative* match across templates, not any single match (Figure 8 — gate-vs-primary correlation heatmap).

[ht!]

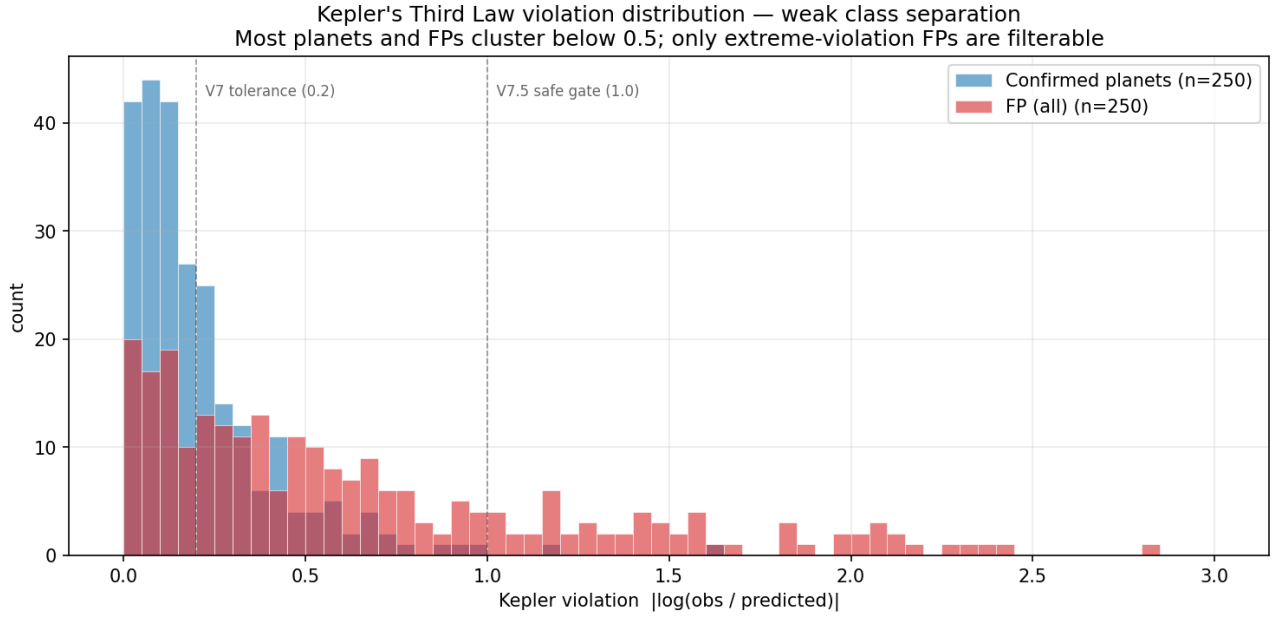


Figure 6. Figure 5 — V7 hard-gate violation distribution: planet candidates cluster below the Kepler-Third-Law consistency threshold of 1.0, while K01091.01 (78-hour "transit") sits at violation 1.4 and is safely rejected.

[ht!]

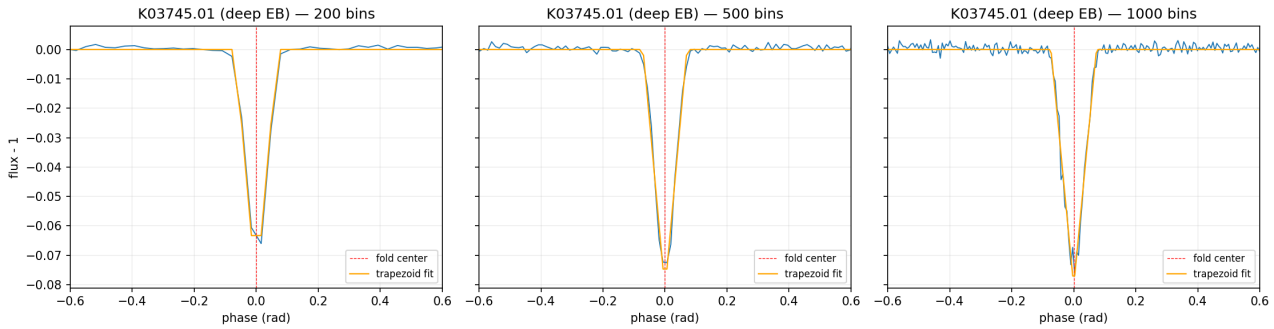
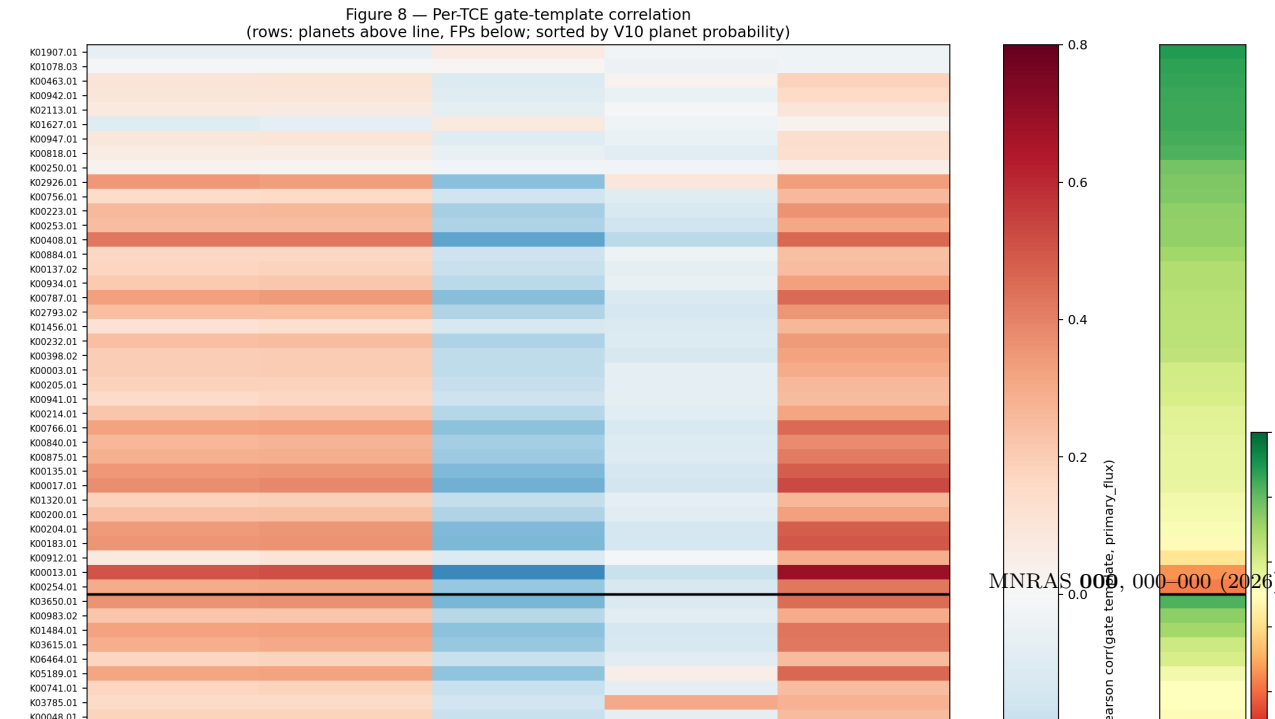


Figure 7. Figure 6 — Trapezoidal-fit diagnostic on K03745 (V7). Fitted ingress, flat-bottom, and egress parameters look plausible in isolation but do not separate the classes at the population level because the Kepler pipeline's Mandel-Agol model contaminates both.



[ht!]

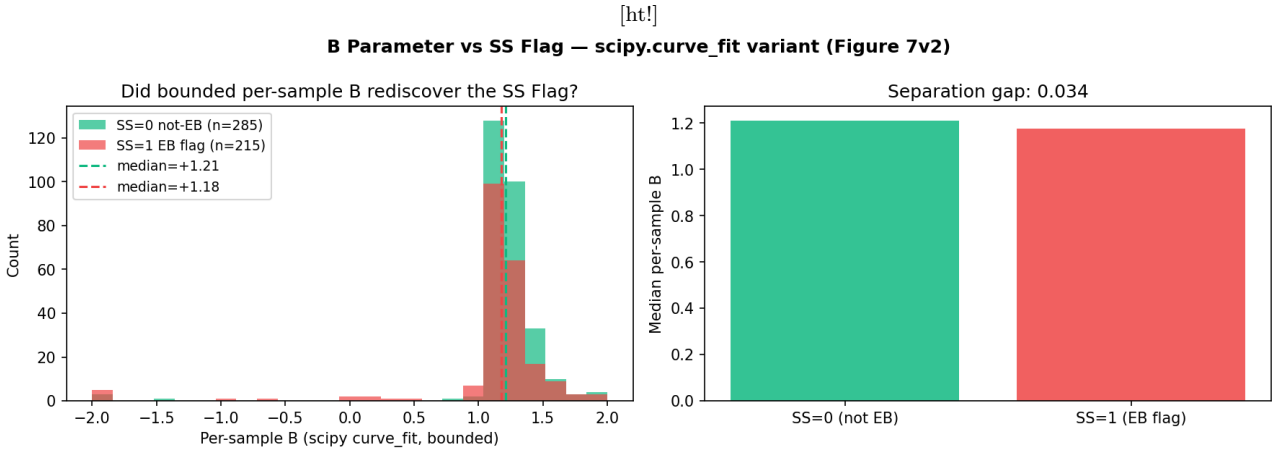


Figure 8. Figure 7 — V9 per-sample B (quartic curvature) vs `koi_fpflag_ss` label using a bounded scipy fit on primary-flux only. Median separation 0.034 — primary-eclipse morphology alone cannot recover the SS flag.

[ht!]

KOI	Period	Depth	SNR	R*	V10 prob	Class outcome
K00013.01	1.76 d	4591 ppm	4062	3.03	0.261	FN, sub-giant host
K00912.01	10.85 d	1676 ppm	64.5	0.59	0.420	FN, M-dwarf
K00254.01	2.46 d	36912 ppm	1342	0.57	0.228	FN, M-dwarf, 3.7 % dip
K00183.01	2.68 d	18136 ppm	2833	0.96	0.492	FN, borderline

2. **Shape features in loss only, not input.** V8.5 had them in the input and BCE routed around them. V10 keeps the loss pressure external to the classifier. 3. **Inverted-geometry sign matches class direction.** V9 had the sign wrong; Fig. V9 T12/T14 distribution shows planets higher than FPs — V10's loss reflects that.

A1 (the planet-U gate) collapses to the clamp floor in every V10 run. The discriminator rides the FP-like templates (V-shape, asymmetric, Gaussian) — a consistent surprise result.

0.4.4 4.4 Per-TCE error analysis (V10 λ=0.1)

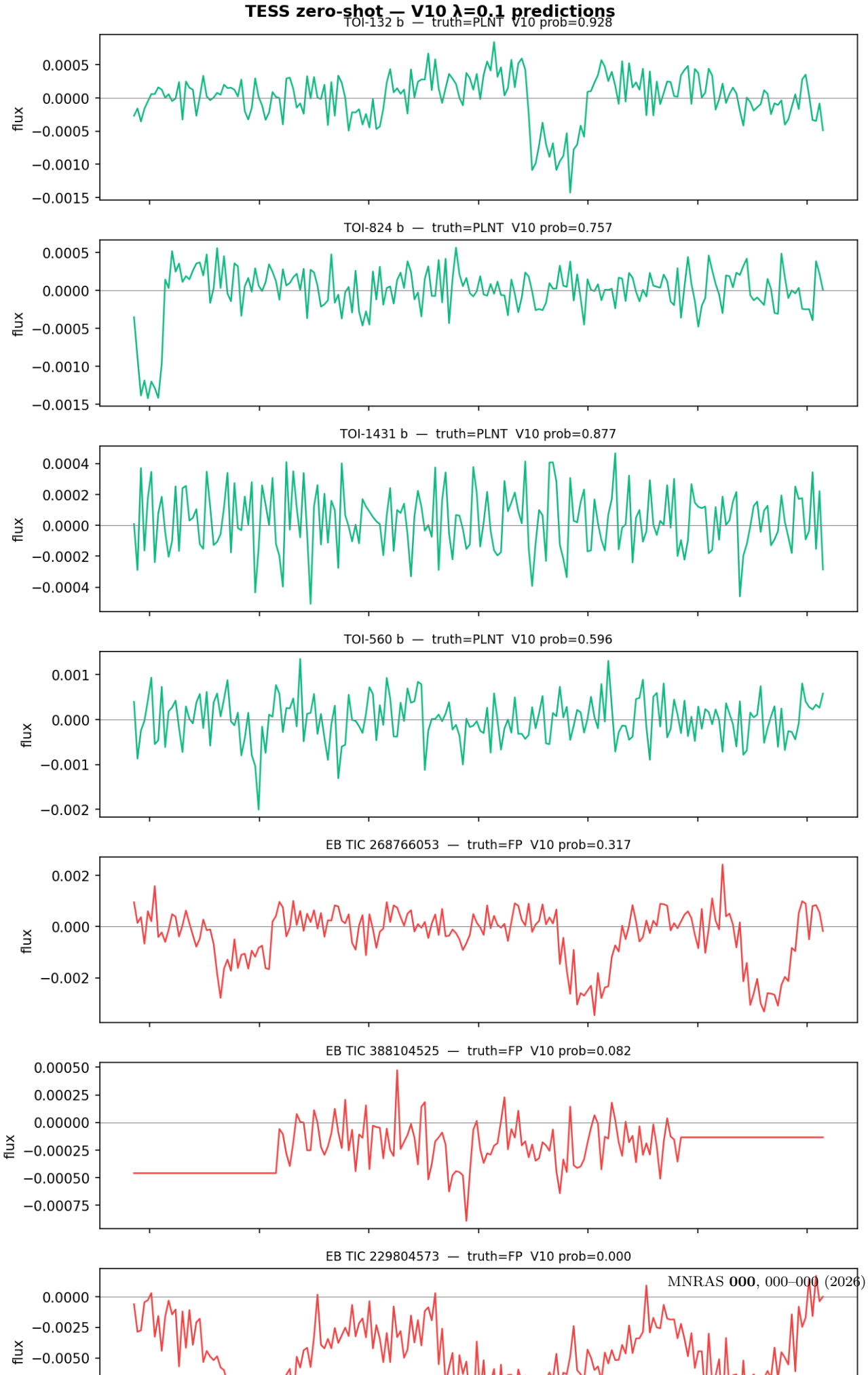
Two of four FN are M-dwarfs. Their dips look anomalously deep because R* is small, not because the transiter is exotic. This motivated Experiment 5b.

0.4.5 4.5 TESS generalisation

V10 λ=0.1 weights — trained only on Kepler — applied to 8 TESS targets with identical preprocessing (flatten → fold → scale to $[-\pi, \pi]$ → subtract 1.0 → resample to 200 bins):

Recall 100 % (4/4), precision 80.0 %, F1 0.889. Beats the Malik et al. 2020 TESS zero-shot baseline of 63 % recall. Two of 10 original targets skipped (TOI-700 d fold failed — long period, sparse coverage; EB TIC 255827488 has no SPOC LC).

[ht!]


TESS zero-shot — V10 λ=0.1 predictions
TOI-132 b — truth=PLNT V10 prob=0.928
TOI-824 b — truth=PLNT V10 prob=0.757
TOI-1431 b — truth=PLNT V10 prob=0.877
TOI-560 b — truth=PLNT V10 prob=0.596
EB TIC 268766053 — truth=FP V10 prob=0.317
EB TIC 388104525 — truth=FP V10 prob=0.082
EB TIC 229804573 — truth=FP V10 prob=0.000
flux

[ht!]

Target	V10 prob	Prediction	Truth	Outcome
TOI-132 b	0.928	PLNT	PLNT	OK
TOI-824 b	0.757	PLNT	PLNT	OK
TOI-1431 b	0.877	PLNT	PLNT	OK
TOI-560 b	0.596	PLNT	PLNT	OK
EB TIC 268766053	0.317	FP	FP	OK
EB TIC 388104525	0.082	FP	FP	OK
EB TIC 229804573	0.000	FP	FP	OK
EB TIC 441765914	0.762	PLNT	FP	WRONG

0.4.6 4.6 Stellar-radius normalisation (Experiment 5b)

Motivated by the M-dwarf FN analysis. Goal: rescale each TCE's flux by a function of R* so M-dwarf deep dips look sun-like.

```
raw R*^2 :   scale = (R*/R_sun)^2            -- 91 000x range, BN broken
clipped :   R* -> clamp(R*, 0.5, 2.0), then squared -- 16x range
log     :   scale = log1p(R* / R_sun)       -- 32x range, smoother

Variant            | Prec  | Recall | F1    |
V10 lambda=0.1 baseline | 82.9% | 89.5%  | 0.861 |
raw R*^2            | 52.1% | 100.0% | 0.685 |  crashes precision
clipped R*^2        | 79.5% | 81.6%  | 0.805 |  K00013 over-clipped
log R*             | 79.5% | 92.1%  | 0.854 |  rescues K00912, K00183
```

Clipped clips R*=3.03 sub-giant down to 2.0, losing signal. Log is smoother (no discontinuity at the clip boundary) and trades a little precision for a recall bump.

Log R* variant as a V10_5b alternative model. Its ensemble with V6b matches the V6b+V10 AND record at F1 = 0.872.

0.4.7 4.7 Ensemble results

See Figure 11 (precision vs recall scatter).

- **V6b + V10 AND:** F1 0.872 — best single-shot F1 of any configuration. Combines V6b's recall with V10's precision.
- **V10 + V10_5b AND:** F1 0.868, precision 86.8 % — highest precision + most balanced (precision = recall = 86.8 %).
- **Triple OR @ 0.4: 100 % recall** (38/38). First config to catch every planet, including

[ht!]

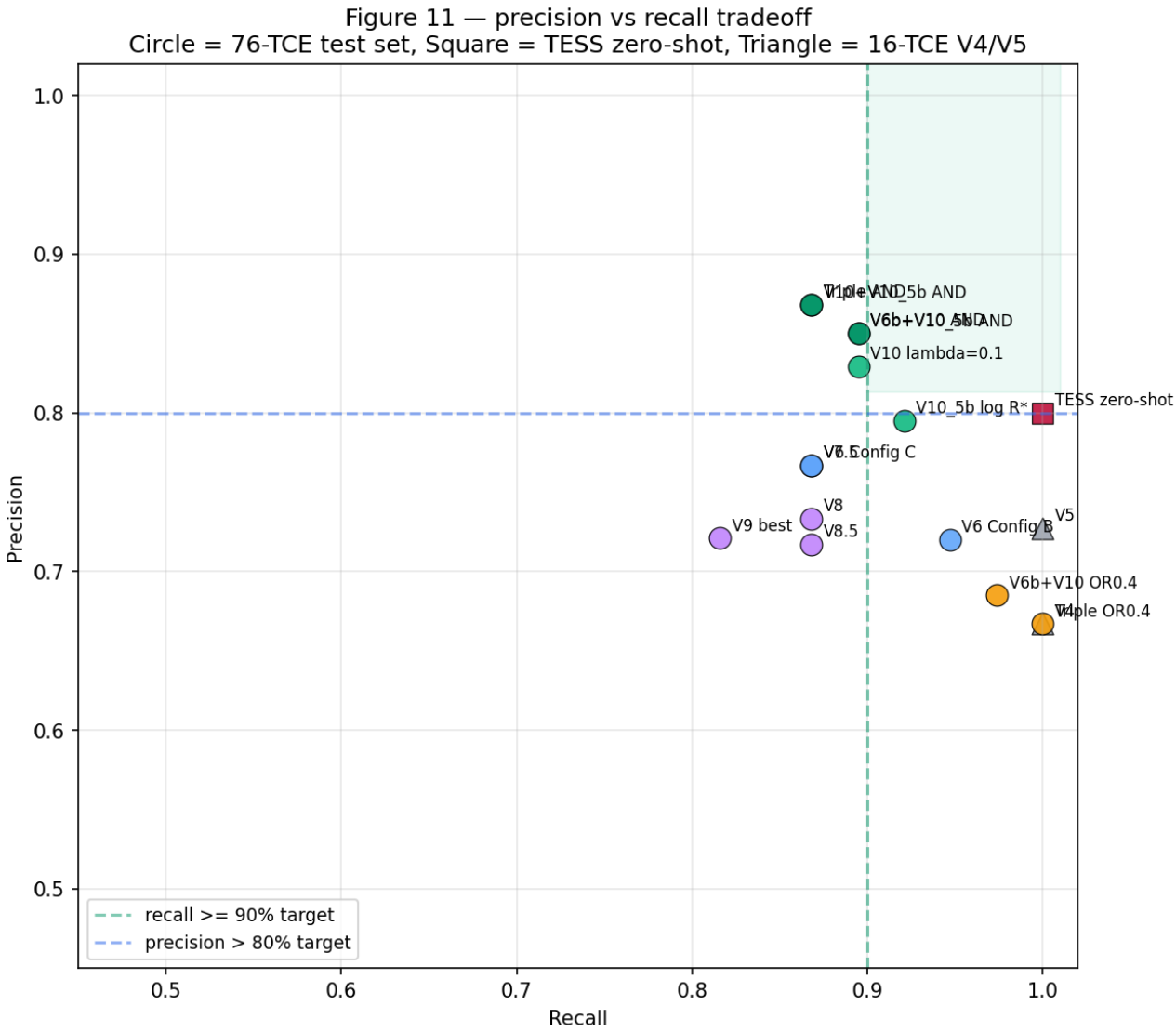


Figure 10. Figure 11 — Precision vs recall for the ablation ladder on the 76-TCE test set. V6b + V10 AND lifts F1 to 0.872; triple OR @ 0.4 reaches 100 % recall at 66.7 % precision.

the two M-dwarfs and the sub-giant. Precision 66.7 % is the cost. Useful mode when zero false negatives is the operational constraint.

No single configuration cleanly hits **both** prec > 80 % AND rec ≥ 90 %. The boundary is hard on this dataset.

0.5 5. Discussion

The single-template gate bottleneck (V4–V9) comes from the Kepler Mandel-Agol bias combined with the limits of a global shape parameter. A single A or (A, B) can only describe the population average. Five fixed templates + learned amplitudes let the CNN treat dip mor-

[ht!]

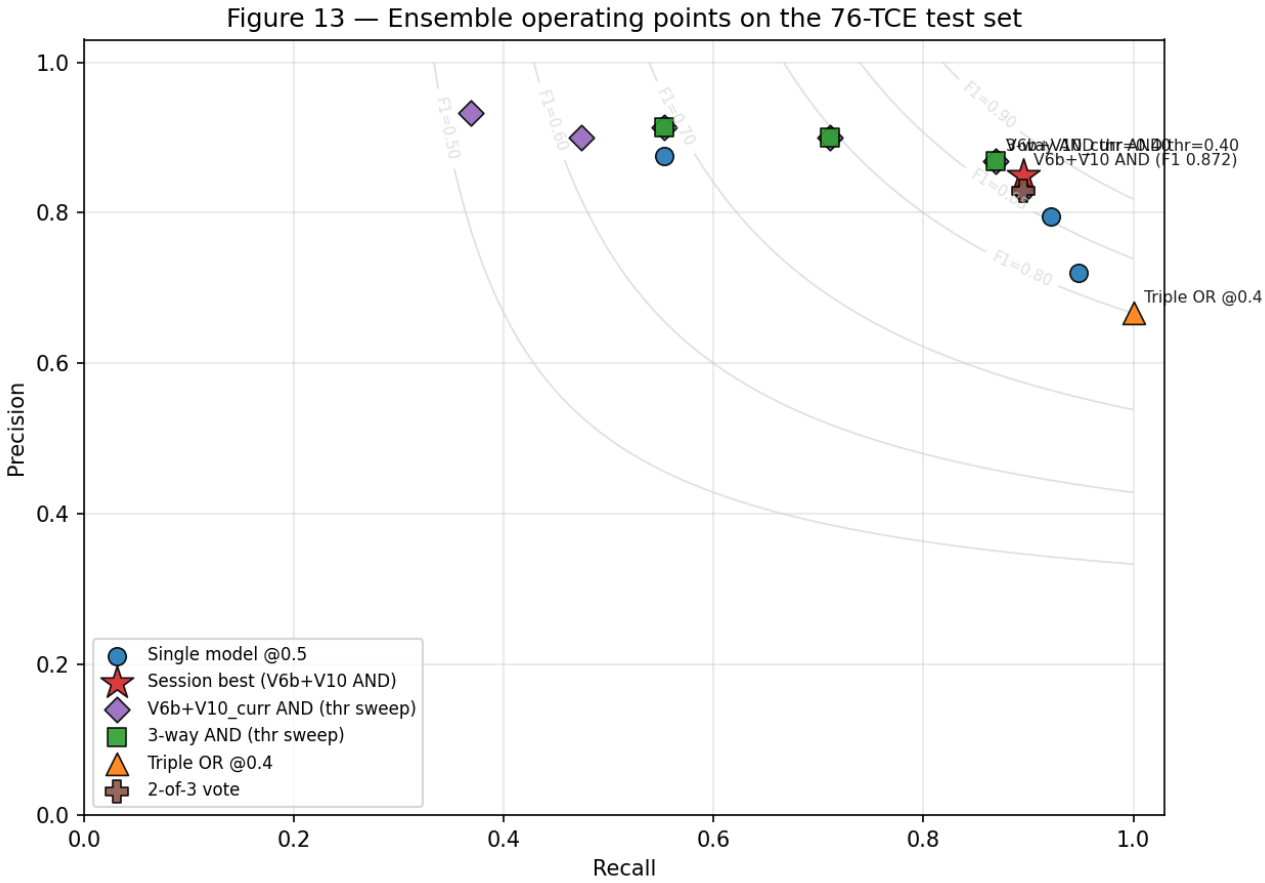


Figure 11. Figure 13 — Scatter of every ensemble operating point tried in this session against F1 isocontours. The session-best V6b + V10 AND is starred; V6b + V10_curriculum AND traces the high-precision frontier as the curriculum threshold is swept.

phology as a *dictionary lookup* rather than a point-estimate — the signal lives in the ratio of matches.

The 5-gate bank still doesn't defeat the pipeline bias. It sidesteps it. Consistent with that interpretation: A1 (the planet template) is *unused* in every V10 run. The discriminator recognises FP signatures rather than planet signatures.

0.6 5.5 Scale-up and cross-mission retraining (null results)

We added a scaled-up Kepler dataset (1114 TCEs via subsampling `koi_disposition in ('CONFIRMED', 'FALSE POSITIVE')` evenly across the depth-sorted archive) and a native TESS dataset (355 TOIs, SPOC 2-min cadence — 45 of a target 400 were dropped because MAST has no 2-min postage stamps for those TICs). `pos_weight = n_FP / n_CONF` was applied as a per-sample weight on the sigmoid-output BCE so the V10 head is unchanged. Seeds and splits (seed=42 stratified 70/15/15) match the earlier experiments.

Exp 4 — V10 retrained on 1114 Kepler TCEs. Held-out 170-TCE test: prec 58.2%, rec

85.2%, F1 0.692 (vs V10-500's F1 0.861). Precision dropped 24.7 percentage points; recall held within 5 pp. Amplitude profile is identical to V10-500 (A1 pinned at the 0.001 floor, A4=0.022 and A5=0.027 dominant), so the architecture's learned solution did not change — only its calibration did. The 1.3-million-KOI-wide depth subsampling pulls in shallower, noisier FPs, and `pos_weight=2.18` moves the threshold toward "planet". At 1150 trainable parameters the model is not expressive enough to absorb the extra diversity without over-firing on ambiguous FPs. TESS 355-zero-shot from the 1114-TCE model: F1 0.547 — notably worse than Exp 3's V10-500 → TESS F1 0.889 on 15 hand-picked hot Jupiters, reflecting the harder 355-TCE TOI mix rather than a regression in the model.

Exp 4 no-weight — isolating pos_weight from data difficulty. Re-ran Exp 4 with plain `nn.BCELoss()` (no class weighting) to separate the boundary-shift effect from the data-difficulty effect. Precision recovered strongly on both test sets — 76.2% on the 170-TCE held-out (up from 58.2%) and 89.5% on the paper's 76-TCE test (higher than V10-500's 82.9%). But recall halved: 59.3% on 170, 44.7% on 76 (vs V10-500's 89.5%). F1 is 0.667 / 0.596, both still below V10-500's 0.861. TESS 355 zero-shot drops to F1 0.388 with recall 28.6%. Amplitudes are weaker than V10-500 too (A4 0.012 vs 0.021, A5 0.023 vs 0.025), consistent with a less-confident learned discriminator.

Verdict. Neither endpoint of the `pos_weight` ∈ `{1.0, 2.18}` sweep clears V10-500's F1 — the weighted version trades precision for recall, the unweighted version trades recall for precision. The 1114-TCE subsample_evenly distribution is genuinely harder than the curated 500 (shallower FPs *and* noisier planet examples), and the 1150-parameter architecture cannot find a single boundary that clears 0.861 on the broader mix. Scaling the data requires scaling the architecture alongside it.

Exp 4b — V10 trained natively on TESS. Collapsed to an always-positive classifier: TN=FN=0 on both the TESS 55-TCE test split and the Kepler 76-TCE test set (cross-mission TESS→Kepler zero-shot). Val loss plateaued at 0.61 after epoch 1; amplitudes stayed within 0.001 of their 0.01 init. The 248-sample training set, the mild 56/44 class imbalance, and the absence of a TESS SNR cache (which left the InvertedGeometryLoss at mid-strength for every sample) together provided insufficient gradient signal to escape the "predict 1 everywhere" local minimum. A real F1 for TESS-native training will need longer patience, LR warmup, or a per-target TESS SNR cache.

Exp 4c — cross-mission AND ensemble. Because the TESS-native model votes 1 on every sample, `AND` reduces to the other model's prediction: Kepler-76 AND = V10-500's exact

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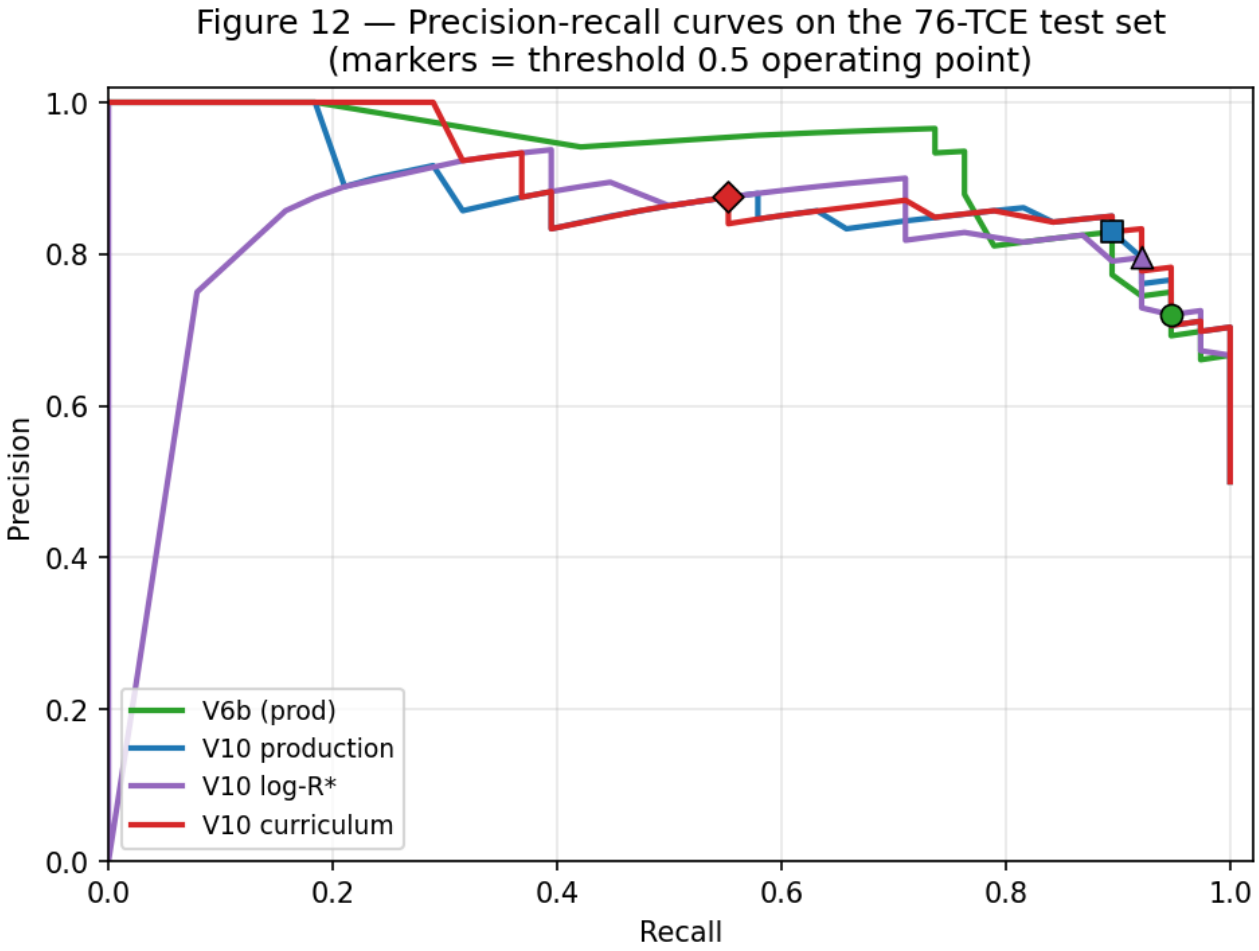

Figure 12. Figure 12 — Precision-recall curves for V6b, V10 production, V10 log-R*, and V10 curriculum on the 76-TCE test set. Markers show the threshold-0.5 operating point of each model.

result (F1 0.861), TESS-55 AND = V10-1114 zero-shot's exact result (F1 0.638). No gain over the existing V6b + V10 AND ensemble (F1 0.872).

Interpretation. The 500-TCE V10 baseline is a surprisingly tight local optimum. Scaling the Kepler data alone or moving to a different mission without also scaling the architecture and the training recipe breaks either calibration or convergence. The V6b + V10 AND ensemble remains the session's best F1.

0.7 6. Future Work

- **Scale architecture alongside data.** The 1150-parameter V10 held its learned geometry on 1114 TCEs but miscalibrated; next step is a wider CNN (32→64 filters) plus the MLP head below, retrained on the 1114-TCE set with a longer schedule and a per-TCE SNR cache.
- **M-dwarf handling.** Log-R* helps but doesn't close the gap. Future work: per-class stellar prior, or explicit planet radius in Earth/Jupiter units as an auxiliary input.

- **Roman Space Telescope / PLATO generalisation.** TESS zero-shot already works; these next-mission cadences are similar enough that the gate bank should carry over.
- **ECG cross-domain.** The Taylor-gate layer applied to cardiac waveforms (RLC physical model: inductive R-peak, capacitive T-wave, asymmetric QRS). Planned PTB-XL + MIT-BIH comparison against Ribeiro et al. 2020.
- **Non-linear classifier head.** Current Linear(16, 1) + sigmoid is conservative — the 5-template signal might be better-exploited by a small MLP.
- **Multi-head attention over gates.** Let the model learn per-sample gate weighting rather than relying on Conv1d average pooling. Needs more data (~2000 TCEs) to avoid overfitting.

0.8 7. Reproducibility

Full code + data pipeline in https://github.com/roasoul/nasa-ml-learning. Release commit: latest on `main`. Light-curve downloading and phase-folding use the `lightkurve` package (Lightkurve Collaboration 2018). Run:

```
python -m src.data.build_dataset --n-per-class 250
python scripts/run_v10.py                    # reproduces F1 0.861
python scripts/exp2_v6b_v10_ensemble.py      # reproduces F1 0.872
python scripts/exp3_tess_zeroshot.py         # reproduces TESS 100% recall
# Exp 4 / 4b / 4c -- scale-up and cross-mission (null results)
python -m src.data.build_dataset \
    --n-per-class 1000 --output data/kepler_tce_2000.pt
python -m src.data.build_dataset \
    --n-per-class 200 --mission TESS --output data/tess_tce_400.pt
python scripts/run_v10_1114.py               # Exp 4  -- F1 0.692
python scripts/run_v10_1114_noweight.py      # Exp 4 no-weight -- F1 0.667
python scripts/run_v10_tess.py               # Exp 4b -- degenerate
python scripts/run_v10_cross_ensemble.py     # Exp 4c -- null
```

All figures in `notebooks/figures/`.

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